

Methods: Java

The body of a Java method

In Java, we use curly brackets `{ }` to enclose the body of a method.

The statements written inside the `{ }` are executed when a method is called.

```
public class Maths {  
    public static void sum(int a, int b) {  
        // Start of sum  
        int result = a + b;  
        System.out.println("Sum is " +  
result);  
    } // End of sum  
  
    public static void main(String [] args)  
{  
        // Here, we call the sum method  
        sum(10, 20);  
        // Output: Sum is 30  
    }  
}
```

Method parameters in Java

In java, parameters are declared in a method definition. The parameters act as variables inside the method and hold the value that was passed in. They can be used inside a method for printing or calculation purposes. In the example, a and b are two parameters which, when the method is called, hold the value 10 and 20 respectively.

```
public class Maths {  
    public int sum(int a, int b) {  
        int k = a + b;  
        return k;  
    }  
  
    public static void main(String [] args)  
{  
        Maths m = new Maths();  
        int result = m.sum(10, 20);  
        System.out.println("sum is " +  
result);  
        // prints - sum is 30  
    }  
}
```

Java Variables Inside a Method

Java variables defined inside a method cannot be used outside the scope of that method.

```
//For example, `i` and `j` variables are  
available in the `main` method only:  
  
public class Maths {  
    public static void main(String [] args)  
{  
        int i, j;  
        System.out.println("These two  
variables are available in main method  
only");  
    }  
}
```

Returning info from a Java method

A Java method can return any value that can be saved in a variable. The value returned must match with the return type specified in the method signature.

The value is returned using the `return` keyword.

```
public class Maths {  
  
    // return type is int  
    public int sum(int a, int b) {  
        int k;  
        k = a + b;  
  
        // sum is returned using the return  
keyword  
        return k;  
    }  
  
    public static void main(String [] args)  
    {  
        Maths m = new Maths();  
        int result;  
        result = m.sum(10, 20);  
        System.out.println("Sum is " +  
result);  
        // Output: Sum is 30  
    }  
}
```

Declaring a Method

Method declarations should define the following method information: scope (private or public), return type, method name, and any parameters it receives.

```
// Here is a public method named sum whose  
return type is int and has two int  
parameters a and b  
public int sum(int a, int b) {  
    return(a + b);  
}
```



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